

# Oracle DBA Cheat Sheet

| Topic                 | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Check Query/Command                                                                                                                                                                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oracle Database       | Robust RDBMS for high-performance data management.                                                                                                                                                                                                                                                                                                                                                                                                                   | SELECT * FROM v\$version;                                                                                                                                                                                                                                                                           |
| RMAN Backup           | Consistent (Cold, offline, MOUNT mode), Inconsistent (Hot, online, ARCHIVELOG mode), Full (all data), Incremental (daily changes).                                                                                                                                                                                                                                                                                                                                   | RMAN> report schema;<br>RMAN> list backup;                                                                                                                                                                                                                                                          |
| RMAN Cold Backup      | SHUTDOWN IMMEDIATE; STARTUP MOUNT; BACKUP DATABASE;                                                                                                                                                                                                                                                                                                                                                                                                                  | SELECT status FROM v\$instance;                                                                                                                                                                                                                                                                     |
| RMAN Hot Backup       | BACKUP DATABASE PLUS ARCHIVELOG; (Requires ARCHIVELOG mode).                                                                                                                                                                                                                                                                                                                                                                                                         | ARCHIVE LOG LIST;                                                                                                                                                                                                                                                                                   |
| Data Guard            | Standby: Clone of production DB. Types: Physical (block-by-block, same DB ID), Logical (SQL apply, separate DB ID), Snapshot (testing with flashback). Processes: LNS (redo transport), MRP (physical apply), LSP (logical apply). Modes: Max Performance, Max Availability, Max Protection.                                                                                                                                                                         | SELECT protection_mode FROM v\$database;<br>SELECT database_role FROM v\$database;<br>SELECT switchover_status FROM v\$database;<br>SELECT process, status FROM v\$managed_standby;                                                                                                                 |
| Data Guard Parameters | DB_NAME, DB_UNIQUE_NAME, LOG_ARCHIVE_DEST_n, FAL_SERVER, STANDBY_FILE_MANAGEMENT, LOG_ARCHIVE_CONFIG.                                                                                                                                                                                                                                                                                                                                                                | SHOW PARAMETER db_name;<br>SHOW PARAMETER db_unique_name;<br>SHOW PARAMETER log_archive_dest;<br>SHOW PARAMETER fal_server;<br>SHOW PARAMETER standby_file_management;<br>SHOW PARAMETER log_archive_config;                                                                                        |
| RAC                   | One database, multiple instances for scalability.                                                                                                                                                                                                                                                                                                                                                                                                                    | SELECT inst_id, instance_name, status FROM gv\$instance;                                                                                                                                                                                                                                            |
| ASM                   | Automatic Storage Management for simplified storage.                                                                                                                                                                                                                                                                                                                                                                                                                 | SELECT name, state FROM v\$asm_diskgroup;                                                                                                                                                                                                                                                           |
| GoldenGate            | Real-time data replication using capture/apply.                                                                                                                                                                                                                                                                                                                                                                                                                      | SELECT capture_name, status FROM dba_capture;<br>SELECT apply_name, status FROM dba_apply;                                                                                                                                                                                                          |
| Oracle Architecture   | <b>Instance:</b> SGA (Database Buffer Cache: stores data blocks; Redo Log Buffer: holds redo entries; Shared Pool: SQL, PL/SQL, data dictionary, library cache; Large Pool: RMAN, parallel execution; Java Pool: Java execution), PGA (session memory for sorting), Background Processes (DBWn: writes buffers; LGWR: writes redo logs; CKPT: updates headers; SMON: recovery; PMON: cleans failed processes). <b>Database:</b> Datafiles, control files, redo logs. | SELECT name, value FROM v\$sga;<br>SELECT component, current_size/1024/1024 MB FROM v\$sga_dynamic_components;<br>SELECT process FROM v\$bgprocess WHERE paddr != '00';<br>SELECT name, value/1024/1024 MB FROM v\$pgastat WHERE name IN ('aggregate PGA target parameter', 'total PGA allocated'); |
| User Process          | Client process initiating SQL requests.                                                                                                                                                                                                                                                                                                                                                                                                                              | SELECT sid, username, program FROM v\$session WHERE type='USER';                                                                                                                                                                                                                                    |

|                         |                                                                                                  |                                                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Server Process          | Executes SQL for user.                                                                           | SELECT sid, spid, program FROM v\$process WHERE<br>addr IN (SELECT paddr FROM v\$session);                  |
| Physical Files          | Datafiles (.dbf), control files (.ctl), redo logs (.log),<br>archive logs, spfile/pfile.         | SELECT name FROM v\$datafile;<br>SELECT name FROM v\$controlfile;<br>SELECT member FROM v\$logfile;         |
| Logical Files           | Tablespaces (8KB default block size), segments, ex-<br>tents, blocks.                            | SELECT tablespace_name, segment_name FROM<br>dba_segments;                                                  |
| File Paths              | Datafiles: /u01/oradata/DBNAME/, Control:<br>/u01/control/, Redo: /u01/redolog/.                 | SELECT file_name FROM dba_data_files;<br>SELECT name FROM v\$controlfile;<br>SELECT member FROM v\$logfile; |
| TNS                     | Transparent Network Substrate (TNSNAMES.ORA).                                                    | tnsping <service_name>                                                                                      |
| DB Link                 | Access remote database objects.                                                                  | SELECT db_link, host FROM dba_db_links;                                                                     |
| Patching                | Resolves bugs and vulnerabilities.                                                               | opatch lsinventory;                                                                                         |
| Hot Backup              | ALTER TABLESPACE tbs BEGIN BACKUP; copy files;<br>ALTER TABLESPACE tbs END BACKUP;.              | SELECT tablespace_name, status FROM dba_<br>tablespaces WHERE status='BACKUP';                              |
| EXPDP (Full Backup)     | expdp system/pwd FULL=Y DIRECTORY=dir<br>DUMPFILE=full_exp.dmp LOGFILE=full_exp.log              | SELECT job_name, operation, state FROM dba_<br>datapump_jobs WHERE operation='EXPORT';                      |
| EXPDP<br>(Schema/Table) | expdp system/pwd SCHEMAS=schema_<br>name DIRECTORY=dir DUMPFILE=schema.dmp<br>LOGFILE=schema.log | SELECT job_name, operation, state FROM dba_<br>datapump_jobs WHERE operation='EXPORT';                      |
| IMPDP                   | impdp system/pwd DIRECTORY=dir DUMPFILE=full_<br>exp.dmp LOGFILE=full_imp.log                    | SELECT job_name, operation, state FROM dba_<br>datapump_jobs WHERE operation='IMPORT';                      |
| Data Pump Tuning        | Use DIRECT=Y, PARALLEL=n, schedule via crontab.                                                  | SELECT job_name, degree FROM dba_datapump_<br>jobs;                                                         |
| Data Pump Directory     | OS: mkdir /path/dir, DB: CREATE DIRECTORY dir<br>AS '/path/dir';                                 | SELECT directory_name, directory_path FROM<br>dba_directories;                                              |
| Network Import          | impdp system/pwd NETWORK_LINK=dblink<br>DUMPFILE=exp.dmp LOGFILE=imp.log                         | SELECT db_link, host FROM dba_db_links;                                                                     |
| Grant Privilege         | GRANT SELECT ON table TO user;                                                                   | SELECT grantee, privilege, table_name FROM<br>dba_tab_privs WHERE grantee='USER';                           |
| Revoke Privilege        | REVOKE SELECT ON table FROM user;                                                                | SELECT grantee, privilege, table_name FROM<br>dba_tab_privs WHERE grantee='USER';                           |
| User Management         | CREATE USER user IDENTIFIED BY pwd;<br>ALTER USER user IDENTIFIED BY new_pwd;<br>DROP USER user; | SELECT username, account_status FROM dba_<br>users;                                                         |
| Role Management         | CREATE ROLE role_name;<br>GRANT SELECT ON table TO role_name;                                    | SELECT role, privilege FROM dba_role_privs<br>WHERE grantee='ROLE_NAME';                                    |

|                       |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Profile Management    | CREATE PROFILE prof LIMIT PASSWORD_LIFE_TIME 180;                                                 | SELECT profile, resource_name, limit FROM dba_profiles WHERE profile='PROF';                                                                                                                                                                                                                                                                                                                                                                 |
| Add Datafile          | ALTER TABLESPACE tbs ADD DATAFILE '/path/file.dbf' SIZE 100M;                                     | SELECT file_name, bytes/1024/1024 MB FROM dba_data_files WHERE tablespace_name='TBS';                                                                                                                                                                                                                                                                                                                                                        |
| Resize Datafile       | ALTER DATABASE DATAFILE '/path/file.dbf' RESIZE 200M;                                             | SELECT file_name, bytes/1024/1024 MB FROM dba_data_files WHERE file_name='/path/file.dbf';                                                                                                                                                                                                                                                                                                                                                   |
| Check Tablespace Size | SELECT tablespace_name, SUM(bytes)/1024/1024 MB FROM dba_data_files GROUP BY tablespace_name;     | SELECT tablespace_name, SUM(bytes)/1024/1024 MB FROM dba_data_files GROUP BY tablespace_name;                                                                                                                                                                                                                                                                                                                                                |
| Alter Tablespace      | ALTER TABLESPACE tbs ADD DATAFILE '/path/file.dbf' SIZE 100M AUTOEXTEND ON;                       | SELECT tablespace_name, file_name, autoextensible FROM dba_data_files WHERE tablespace_name='TBS';                                                                                                                                                                                                                                                                                                                                           |
| SQL Utility           | Clear SQL terminal: cl scr                                                                        | cl scr                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| DBA Handover Checks   | Check alert log, backups, tablespace space, listener, RAC health, Data Guard sync, job schedules. | SELECT * FROM v\$alert_log;<br>SELECT destination, status FROM v\$archive_dest;<br>SELECT tablespace_name, (SUM(bytes)/1024/1024) - (SUM(free_space)/1024/1024) MB_USED<br>FROM dba_data_files JOIN dba_free_space<br>USING(tablespace_name) GROUP BY tablespace_name;<br>lsnrctl status;<br>SELECT inst_id, status FROM gv\$instance;<br>SELECT sequence#, applied FROM v\$archived_log;<br>SELECT job_name, state FROM dba_scheduler_jobs; |

## Oracle Files and Paths

### Oracle Files Explained

| File Type      | Extension       | Details                                                                    | Analogy                    |
|----------------|-----------------|----------------------------------------------------------------------------|----------------------------|
| Data File      | .dbf            | Stores tables, indexes, temporary/undo blocks, materialized views, LOBs.   | Notebook with actual data. |
| Redo Log File  | .log            | Sequential log of DML changes (insert/update/delete), transaction vectors. | CCTV live recording.       |
| Archived Log   | .arc            | Backup of redo logs for media/point-in-time recovery.                      | CCTV backup hard drive.    |
| Control File   | .ctl            | DB structure, SCN, datafile names, redo log sequence, backup info.         | Books table of contents.   |
| Parameter File | spfile/init.ora | DB startup config: DB_NAME, memory sizes, paths.                           | Cars ignition settings.    |
| Trace File     | .trc            | Errors, diagnostics, session debug info.                                   | Doctors patient report.    |
| Alert Log      | .log            | DB events: startup/shutdown, errors, tablespace issues.                    | Hospital daily log.        |

### Summary Table

| File            | Data Type            | Written By       |
|-----------------|----------------------|------------------|
| .dbf            | Table data           | DBWR             |
| .log (redo)     | Transaction logs     | LGWR             |
| .arc            | Redo log backup      | ARCH             |
| .ctl            | Metadata & structure | Oracle engine    |
| spfile/init.ora | Startup parameters   | DBA or Oracle    |
| .trc            | Debug info           | Oracle processes |
| alert.log       | DB-wide event log    | Oracle instance  |

### Oracle DBA Important Paths

| Purpose         | Path / File Location                                | Description                                     |
|-----------------|-----------------------------------------------------|-------------------------------------------------|
| Oracle Base     | \$ORACLE_BASE (/u01/app/oracle)                     | Root folder for Oracle software, logs, configs. |
| Oracle Home     | \$ORACLE_HOME (/u01/app/oracle/product/19.0.0/db_1) | Oracle software installation location.          |
| Listener Config | \$ORACLE_HOME/network/admin/listener.ora            | Listener port, host configuration.              |

|                   |                                                                 |                                          |
|-------------------|-----------------------------------------------------------------|------------------------------------------|
| TNS Config        | \$ORACLE_HOME/network/admin/tnsnames.ora                        | Connection strings to DBs.               |
| SQL*Net Config    | \$ORACLE_HOME/network/admin/sqlnet.ora                          | Network protocols, encryption settings.  |
| Listener Log      | /u01/app/oracle/diag/tnlsnr/<hostname>/listener/alert/alert.log | Listener connection attempts and errors. |
| Listener Trace    | /u01/app/oracle/diag/tnlsnr/<hostname>/listener/trace/trace.log | Debug log for listener.                  |
| DB Alert Log      | /u01/app/oracle/diag/rdbms/<db>/<inst>/alert/alert_<inst>.log   | DB errors, startup/shutdown, events.     |
| Datafiles         | SELECT name FROM v\$datafile;                                   | Actual data file paths.                  |
| Redo Log Files    | SELECT member FROM v\$logfile;                                  | Redo log file paths.                     |
| Control Files     | SELECT name FROM v\$controlfile;                                | Control file paths.                      |
| Archive Log Files | /u01/app/oracle/fast_recovery_area/<db_name>/archivelog/        | Archived redo logs.                      |
| Audit Logs        | \$ORACLE_BASE/admin/<db_name>/adump/                            | User activity logs.                      |
| RMAN Backup       | /u01/app/oracle/backup/ or FRA                                  | DB backups via RMAN.                     |
| Background Trace  | /u01/app/oracle/diag/rdbms/<db>/<inst>/trace/                   | Trace files for SMON, PMON, DBWn.        |
| ORATAB File       | /etc/oratab                                                     | Lists all DB instances on machine.       |
| Password File     | \$ORACLE_HOME/dbs/orapw<ORACLE_SID>                             | SYSDBA remote login.                     |

# Oracle DBA Responsibilities and Utilities

## Linux Commands for Oracle DBA

| Command                                       | Purpose                                             |
|-----------------------------------------------|-----------------------------------------------------|
| <code>netstat -tulnp</code>                   | Show listening ports with process info (TCP + UDP). |
| <code>ss -tulnp</code>                        | Faster alternative to <code>netstat</code> .        |
| <code>lsof -i :1521</code>                    | Check process using port 1521.                      |
| <code>ps -ef   grep tnslnr</code>             | Check if Oracle listener is running.                |
| <code>lsnrctl status</code>                   | Show listener status, port, services.               |
| <code>systemctl status firewalld</code>       | Check firewall status.                              |
| <code>firewall-cmd --list-all</code>          | Show open ports in firewalld.                       |
| <code>iptables -L -n -v</code>                | List iptables rules.                                |
| <code>hostname -I</code> or <code>ip a</code> | Check system IP address.                            |
| <code>curl -v</code>                          | Test port 1521 connectivity.                        |
| <code>telnet://localhost:1521</code>          |                                                     |
| <code>telnet localhost 1521</code>            | Check if port accepts connections.                  |
| <code>nc -zv 127.0.0.1 1521</code>            | Check port status using netcat.                     |
| <code>whoami</code>                           | Verify current user (e.g., oracle).                 |

## DBA Responsibilities

| Task               | Details                                                 | Commands                                                                                                                                                                        |
|--------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DB Installation    | Install Oracle software, create DB, configure listener. | <code>./runInstaller -silent -responseFile db_install.rsp</code><br><code>dbca -silent -createDatabase -gdbName ORCL</code><br><code>lsnrctl start</code>                       |
| User Management    | Create users, assign roles/privileges, manage profiles. | <code>CREATE USER app_user IDENTIFIED BY pwd;</code><br><code>GRANT SELECT ON table TO app_user;</code><br><code>CREATE PROFILE app_profile LIMIT PASSWORD_LIFE_TIME 90;</code> |
| Performance Tuning | Optimize queries with indexes, stats, memory tuning.    | <code>EXPLAIN PLAN FOR SELECT ...;</code><br><code>CREATE INDEX idx ON table(col);</code><br><code>EXEC DBMS_STATS.GATHER_TABLE_STATS('SCOTT', 'TABLE');</code>                 |
| Backup & Recovery  | Full/incremental backups, restore/recover data.         | <code>RMAN&gt; BACKUP DATABASE PLUS ARCHIVELOG;</code><br><code>RMAN&gt; RESTORE DATAFILE '/path';</code><br><code>RMAN&gt; RECOVER DATABASE;</code>                            |

|                     |                                                        |                                                                                                                                                      |
|---------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security Management | Strong passwords, auditing, TDE for encryption.        | ALTER SYSTEM SET ENCRYPTION KEY IDENTIFIED BY "pwd";<br>AUDIT SELECT ON table BY ACCESS;<br>ALTER USER user ACCOUNT LOCK;                            |
| Monitoring          | Check alert log, CPU, memory, disk space, AWR reports. | tail -f alert_ORCL.log<br>SELECT name, value FROM v\$sysstat WHERE name LIKE '%CPU%';<br>BEGIN DBMS_WORKLOAD_REPOSITORY.CREATE_-<br>SNAPSHOT(); END; |

## Startup & Shutdown

| Command                                                                                                                                                                                                                   | Description                                                                                                                                                                                                                                      | Use Case                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STARTUP;<br>STARTUP NOMOUNT;<br>STARTUP MOUNT;<br>STARTUP FORCE;<br>SHUTDOWN NORMAL;<br>SHUTDOWN IMMEDIATE;<br>SHUTDOWN ABORT;<br>lsnrctl<br>status/start/stop<br>SELECT INSTANCE_-,<br>NAME, STATUS FROM<br>V\$INSTANCE; | Full DB start (Instance + Mount + Open).<br>Start instance only.<br>Mount DB without opening.<br>Force restart DB.<br>Wait for users to log off.<br>Disconnect users immediately.<br>Force shutdown.<br>Manage listener.<br><br>Check DB status. | Normal startup.<br>Create DB, restore control files.<br>Media recovery, rename datafiles.<br>Recover from issues.<br>Graceful shutdown.<br>Safe shutdown.<br>Last resort.<br>Check or control listener.<br><br>Verify DB state. |

## Alert Log Checks

- **Startup/Shutdown:** tail -f /u01/app/oracle/diag/rdbms/oradb/oradb/alert/alert\_oradb.log
- **Listener Issues:** Check for ORA-12541: TNS:no listener.
- **Errors:** Look for ORA-00600, ORA-01578, etc.
- **Tablespace Issues:** Check ORA-01653 for space errors.
- **Archiver Status:** Verify ARC0: Completed archiving.
- **Checkpoint/Recovery:** Look for Media Recovery Start.
- **Background Failures:** Check for SMON, DBWO errors.
- **Parameter Changes:** Look for ALTER SYSTEM logs.
- **PDB Status:** Check Pluggable database PDB1 opened.